

SIMATS ENGINEERING
Department of Computing Technologies

Problem Statement

Practice Set - Search & Game-Playing Algorithm Dry Runs (Companion to Assessment Tool 2, Q1 & Q5)

Course Name	Artificial Intelligence
Course Code	CSA17
Related Course Outcome	CO2 - Search & Game-Solving Techniques (BL3)
Questions Covered	Q1: A* Search Q2: Minimax with Alpha-Beta Pruning
Question Type	Algorithm Dry Run (step-by-step trace)

Declaration: I certify that the responses submitted against this problem statement represent my original work, and that I have adhered to the guidelines specified for this assessment, in accordance with the Code of Conduct in Annexure A.

Name / Reg. No.: _____

Signature of the Student: _____

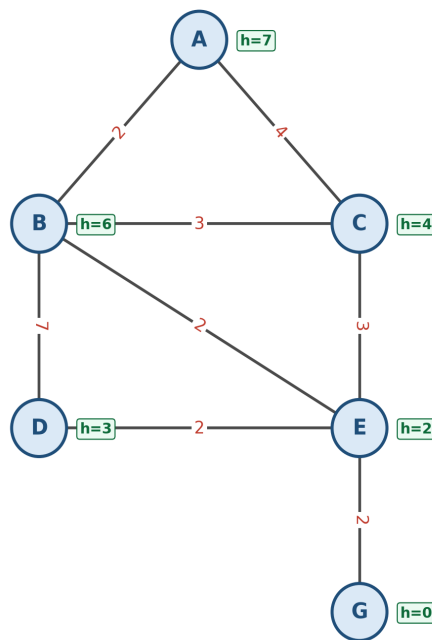
Question 1: Dry Run of A* Search Algorithm

[Total Marks: 10]

1.1 Problem Description

Consider the weighted, undirected graph shown below. Each edge is labelled with its path cost, and every node is annotated with its heuristic estimate $h(n)$ - the estimated cost from that node to the goal node G. Perform a manual dry run of the A* Search algorithm to find a path from the start node A to the goal node G.

Search Graph with Edge Costs and Heuristic Values $h(n)$



1.2 Edge Costs $g(n)$

Edge	Cost
A - B	2
A - C	4
B - C	3
B - D	7
B - E	2
C - E	3
D - E	2
E - G	2

1.3 Heuristic Values $h(n)$

Node	$h(n)$
A	7
B	6
C	4
D	3
E	2
G	0

1.4 Task

Perform a dry run of the A* Search algorithm from start node **A** to goal node **G**. For each iteration of the algorithm, clearly show the following:

- The current node selected for expansion
- The Open List (all frontier nodes, with their f-values)
- The Closed List (all fully expanded nodes)
- $g(n)$ - the exact path cost so far from the start node
- $h(n)$ - the heuristic estimate to the goal
- $f(n) = g(n) + h(n)$ - the evaluation function value

Finally: (a) identify the optimal path from A to G found by the algorithm, and (b) calculate its total path cost.

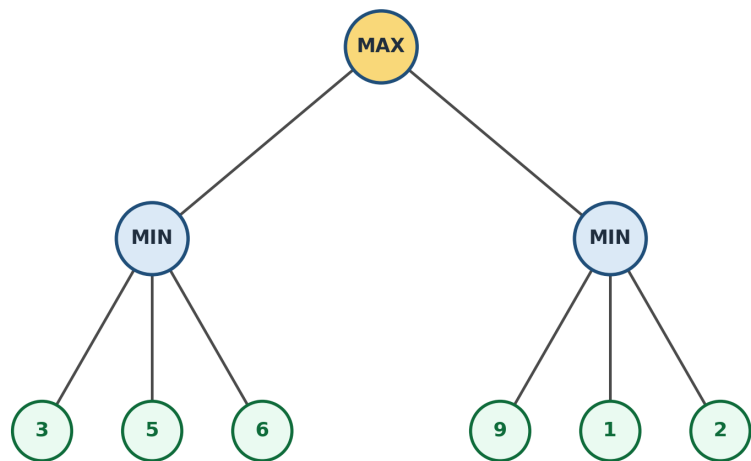
Question 2: Dry Run of Minimax with Alpha-Beta Pruning

[Total Marks: 10]

2.1 Problem Description

Consider the game tree shown below. The root and all internal nodes at the second level alternate between MAX and MIN, and the leaf nodes hold the terminal utility values. Perform a manual dry run of the Minimax algorithm with Alpha-Beta Pruning, evaluating the tree strictly left to right.

Game Tree for Minimax with Alpha-Beta Pruning



2.2 Task

Evaluate the tree from left to right using Alpha-Beta Pruning. For each node, show:

- Alpha (α)
- Beta (β)
- Selected value
- Any node that gets pruned

Finally: (a) determine the best move for MAX, (b) state the final minimax value, and (c) list the pruned nodes (if any).

2.3 Marking Scheme (Reference)

Component	Marks
Correct alpha/beta trace at every node	5
Correct identification of pruned node(s) with justification	2
Correct best move for MAX	1
Correct final minimax value	2
Total	10